



Improving the Hurricane Outage Prediction Model by including tree species



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ABSTRACT

Hurricanes can be a major threat to electrical power systems, often resulting in costly repairs and lengthy restoration times. The Hurricane Outage Prediction Model (HOPM) predicts the location and number of outages for a major coastal Gulf of Mexico utility company up to 5 days before a hurricane makes landfall. This model can help electrical utilities to improve their resource allocations and potentially shorten restoration times. The goal of this study is to evaluate whether the accuracy of the HOPM can be improved by including information on tree species. Our results demonstrate that the model accuracy increased by ~3% when tree species were included. The most important tree species in our study region were Sweetgum and Loblolly Pine. As the relative abundance of these tree species increased, the number of outages tended to increase. In contrast, increases in the relative abundance of Water Oak and Chestnut Oak did not result in more outages. This suggests that certain oaks may be more resistant to uprooting or snapping when there are strong winds. Therefore, the inclusion of tree species variables in the model provides a means for capturing the spatially varying vulnerability of the trees to strong winds. Even though including tree species improved the accuracy of the model, information about the frequency of tree trimming is more important. Accurate prediction of the location and number of power outages requires information on both the meteorological hazard and the vulnerability of the vegetation.

1. Introduction

Tropical cyclones (known as hurricanes in the Western Atlantic and Eastern Pacific Ocean Basins) are one of the deadliest and most destructive natural disasters (Emanuel, 2005). Violent landfalling hurricanes will often damage homes, businesses, and infrastructure. This causes major interruptions to livelihoods and day-to-day operations. Hurricanes can also render significant damage to electric power distribution systems, causing major disruptions to service. For example, during Hurricane Katrina in 2005, 82% of customers in one of the largest utility companies along the United States Gulf Coast lost power and some customers waited up to 12 days to have their power restored (Guikema et al., 2010). Disruptions to power systems, such as those cause by Hurricane Katrina, can have profound impacts on other lifelines, like hospitals, water delivery and sanitation, and emergency services (Nojima and Sugito, 2002; Chang et al., 2004) including the delivery of energy on regional and national scales (Reed et al., 2010a) and the risk of and recovery from hygiene, illness, and disease issues (Vanderford et al., 2007). Therefore, predicting hurricane impacts on the power

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grid prior to landfall can improve pre-storm planning efforts for utility companies.

Electrical utilities generally lack sufficient personnel to rapidly restore power after major hurricanes. Instead, through mutual aid agreements, they rely on other utilities for assistance. Therefore, they need to estimate the number of additional crews that they will require to restore power quickly and efficiently. Requesting more assistance than necessary will result in unnecessary expenditures, while requesting too little assistance will result in significant delays in power restoration. As part of their pre-storm planning efforts, utilities must weigh the anticipated impacts from the storm and make an estimate on how much assistance they will need (Guikema et al., 2010).

Models to forecast weather-related power outages typically use many predictors to forecast outages or damage to electrical infrastructure. The predictors that are common to these models include: storm conditions, power system characteristics, power system reliability, and environmental characteristics such as land cover. These models generally utilize power system performance data from past hurricanes to predict the potential impacts from an approaching hurricane (Guikema et al., 2010). These predictions enable crews to be located prior to the storm in the areas of greatest need. It also allows for a better estimate of the additional resources the utility will require to restore power quickly and efficiently.

The two primary ways of modeling the impact of hurricanes on the electrical power system are fragility-based approaches (Davidson et al., 2003; Reed et al., 2010b; Winkler et al., 2010; Ouyang and Dueñas-Osorio, 2014) and statistical/machine learning approaches. This paper uses the latter approach because it tends to be more flexible, accurate and faster.

Krishnamurthy and Kwasinski (2013) used a non-linear regression model to generate county-level predictions of the maximum number of outages, average outage duration, and restoration time. Their outage predictions were more accurate than their restoration predictions, potentially related to regional variance in restoration practices. Storm surge, maximum sustained wind speed, storm size, and the exposure time to tropical storm force winds were all found to be important predictor variables, with storm surge being the strongest predictor in coastal areas (Krishnamurthy and Kwasinski, 2013).

Negative binomial regression models have also proven effective for modeling hurricane-induced power outages. Liu et al. (2005) employed such a model to predict the number of outages at the zip code level. They found that negative binomial regression yields more accurate predictions than Poisson regression. The most influential variables in their models were the number of transformers, the company affected, maximum wind gust, and a hurricane indicator variable. Soil drainage may also be important, but only when trees are located close to power lines (Liu et al., 2005).

Liu et al. (2008) developed a spatial generalized linear mixed model (GLMM) to predict hurricane-related power outages. Their model tended to over-predict outages due to hurricanes. The number of protective devices and the maximum wind gust speed were the two variables that had the greatest impact on hurricane-induced power outages.

Previous studies have found that vegetation-related variables such as vegetation management/tree trimming and vegetation abundance are important for predicting power outages. Simpson and Van Bossuyt (1996) determined that tree failure was the primary cause for 40% of outages and suggested that many of these outages could be prevented by tree trimming and maintenance around overhead lines. In a study involving Duke Power, Guikema et al. (2006) examined power outages in non-storm conditions and concluded that tree trimming can greatly reduce the number of outages in these conditions. Reed (2008) examined power outages following the Winter Storm of 1999 in France and winter storms in Seattle and found that many of the outages were caused by trees falling on power lines. Maliszewski et al. (2012) stated that the interactions between birds, vegetation and overhead lines were the second most important factor in predicting distribution system outages in Phoenix, AZ. Tree trimming has also been identified as an important variable in hurricane power outage models. Nateghi et al. (2014) found that tree trimming was one of the six most important predictors of hurricane-related power outages.

This study utilizes the Hurricane Outage Prediction Model (HOPM). HOPM is a statistically-based model which utilizes hurricane, environmental and power system predictors to estimate the quantity and spatial distribution of power outages prior to hurricane landfall. HOPM was initially developed by Han et al. (2009b) using a negative binomial regression model. HOPM has evolved over time to include more variables and it is now based on Random Forest (Han et al., 2009a,b; Quiring et al., 2011; Nateghi et al., 2014; Guikema et al., 2014). The version of HOPM that is being used in this study is from Nateghi et al. (2014). While this model includes a measure of tree trimming frequency, it does not include any other tree-related variables. Past studies have noted that one of the major limitations of power outage models is the lack of tree-related predictor variables, such as the number of trees, tree species, age and tree height (Liu et al., 2008).

Davidson et al. (2003) concluded that tree species are an important predictor for hurricane related power outages because some species are more prone to limb breakage, while other species have shallower rooting systems and therefore are more prone to uprooting. Using data from Duke Power during Hurricanes Opal, Fran, and Floyd, Davidson et al. (2003) found that trees accounted for about half of the power outages.

Duryea et al. (2007) studied the impact of winds from eight hurricanes on urban tree species in Florida, each making landfall between 1992 and 2005. They found that certain tree species were more likely to survive hurricane force winds. The trees that did not survive the hurricanes succumbed to branch loss, uprooting, or snapping, depending on the species of the trees (Duryea et al., 2007). Ultimately, Duryea et al. (2007) compiled a ranking of tree species based on their wind resistance. Table 1 compares the characteristics and survival rates of 9 tree species based on data from six hurricanes (Gresham et al., 1991; Barry et al., 1993; Duryea et al., 2007; Xi and Peet, 2008) which affected forested regions of the southeastern United States. The tree species that experienced less breakage and uprooting, and a higher amount of defoliation generally had a higher rate of survival (Duryea et al., 2007).

This study will evaluate whether the inclusion of tree-species data improves the accuracy of the Hurricane Power Outage Model. Specifically, it will determine the relative importance of various tree species and how they impacted outage predictions for four hurricanes that made landfall in the U.S. Gulf Coast region.

Table 1

Tree species characteristics vs. hurricane survival. For breakage and uprooting, the species less likely to break or uproot (more likely to survive) are listed from top to bottom. For defoliation, the species more likely to defoliate (more likely to survive) are listed from top to bottom. The survival percentages are from [Duryea et al. \(2007\)](#) and are based on six hurricanes.

Breakage	Uprooting	Defoliation	Survival
Sweetgum (less)	Sweetgum (less)	Dogwood (more)	Dogwood (92%)
Dogwood	White Oak	Sweetgum	Sweetgum (90%)
Water Oak	Chestnut Oak	Laurel Oak	Laurel Oak (87%)
White Oak	Longleaf Pine	Loblolly Pine	Slash Pine (85%)
Chestnut Oak	Slash Pine	Water Oak	Longleaf Pine (75%)
Virginia Pine	Loblolly Pine	Longleaf Pine (less)	Loblolly Pine (74%)
Longleaf Pine	Water Oak		Water Oak (72%)
Slash Pine	Dogwood (more)		
Loblolly Pine (more)			

2. Data and methods

2.1. Study region

Data for this study were provided by a major Gulf Coast utility who wishes to remain anonymous. The region used for this study covers one state that is served by this utility. The utility has divided their service area into grid cells that are 3.66 km (12000 ft) by 2.44 km (8000 ft) ([Quiring et al., 2014](#)). State A consists of 6681 grid cells, which covers approximately 59,615 km² ([Fig. 1](#)). Gaps in grid cell coverage indicate areas lacking people or areas covered by another utility provider. State A was selected for this study to facilitate comparison with the results of [Nateghi et al. \(2014\)](#) and because it was strongly affected by three of the four hurricanes used in this study ([Fig. 1](#)).

2.2. Hurricanes

This study is based on data from four hurricanes: Danny ([Pasch, 1997](#)), Ivan ([Stewart, 2004](#)), Dennis ([Beven, 2005](#)), and Katrina ([Knabb et al., 2005](#)). All four hurricanes tracked through the State A service area ([Fig. 1](#)). Danny formed near the coast of LA and moved slowly east-northeastward, eventually making landfall near the Mississippi River delta as a Category 1 hurricane ([Schott et al., 2012](#)). A second landfall occurred near the mouth of Mobile Bay on July 19th, 1997. Following its second landfall, Danny continued to meander slowly northward for several days, gradually weakening to a tropical depression ([Pasch, 1997](#)). Ivan originated off the west coast of Africa and reached Category 5 strength in the Caribbean. However, it weakened to a Category 3 storm just prior to landfall near Gulf Shores, AL on September 16th, 2004 and by September 17th, Ivan had weakened to a tropical depression ([Stewart,](#)

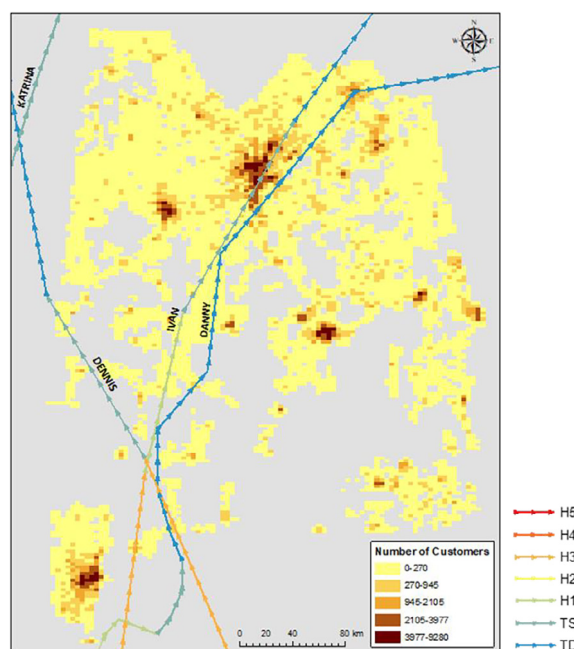


Fig. 1. The study area in State A showing the number of customers and the tracks of the four hurricanes used in this study.

2004). Dennis also originated off the west coast of Africa and eventually made landfall in Cuba as a Category 4 hurricane. After departing Cuba, Dennis continued its trek westward and northwestward, ultimately making landfall on Santa Rosa Island, just east of Pensacola on the Florida Panhandle, as a Category 3 storm (Beven, 2005). Katrina developed over the Bahamas and made its first landfall on the southeastern FL peninsula as a Category 1 storm. As Katrina reemerged over the Gulf of Mexico, it rapidly strengthened into a Category 5 hurricane as it approached the Central Gulf Coast. A second landfall was made along the Mississippi Delta near Buras, LA on August 29th, 2005 as a strong Category 3 storm, followed by a final landfall near the LA/MS state line. Katrina is the strongest and most destructive hurricane considered in this study, with significant impacts extending hundreds of miles from the center (Knabb et al., 2005).

2.3. Tree species data

The host-tree species dataset from the 2012 National Insect and Disease Risk Map (NIDRM) is used in this study. This dataset was developed by the Forest Health Technology Enterprise Team (FHTET) of the U.S. Department of Agriculture (USDA) Forest Service to model tree species risk due to insects, diseases, and other hazards (Krist et al., 2014). This dataset was chosen for use in this study because it provides comprehensive coverage of the United States, including urban areas. The number and grouping of species classes are also an improvement over the 2008 U.S. Forest Type dataset (Ruefenacht et al., 2008), which, to our knowledge, is the only other tree species dataset that provides complete coverage of the United States.

The 2012 NIDRM host species dataset was derived by predicting tree species parameters (i.e., basal area, stand density index, quadratic mean diameter) for 300,000 USDA Forest Service Forest Inventory and Analysis (FIA) plots using a classification and regression trees (CART) model. The model included FIA plot data that were overlaid with a wide variety of geospatial predictor layers, including soils, climate, terrain, and Landsat variables. The 2012 NIDRM is a dramatic improvement over the 2006 NIDRM which was based solely on FIA data and used a simple Inverse Distance Weighting (IDW) technique, instead of a regression-based model, to determine tree species (Krist et al., 2014).

The 2012 NIDRM provides tree species data at a 240-meter resolution. These data are re-sampled to the resolution of the utility company grid cells using Zonal Statistics in ArcGIS. The majority (most prevalent) tree species was assigned to each utility grid cell. A better approach for isolating the impact of tree species on power outages would be to identify the trees that are within a given distance from the overhead power lines. However, the utility that we are working with did not provide the location of the overhead lines, only the total length of overhead lines in each grid cell. Therefore, we are assuming that the most prevalent tree species in the grid cell represents that trees that are close to the power lines. This is one of the sources of uncertainty in this study. All tree species that were identified as the majority tree species in at least 0.5% of the grid cells in State A were retained for this analysis. This resulted in 12 NIDRM classes: 10 of these classes are shown in Table 2 and Fig. 2; one is called “Other” and it refers to all the other tree species that are not a majority species in at least 0.5% of the study region; and the final class is called “Empty”, representing locations where data are missing. The Other class and the Empty class were combined into one class. A comparison with the 2011 National Land Cover Dataset (<http://www.mrlc.gov/nlcd2011.php>) confirmed that the areas covered by this combined class contain little to no tree cover. These 11 classes account for 98% of the grid cells within the service area (Table 2). The remaining 2% are tree species that do not have enough areal coverage to be considered significant in the State A study region. An additional source of uncertainty in our approach is assigning the most prevalent tree species to each grid cell as opposed to considering the relative abundance of each tree species within each grid cell. This approach was taken to avoid adding dozens of new variables to the model, many that would generally have very small values. However, there are some locations where there is substantial variability in tree species within a single grid cell and this will not be captured in our approach.

2.4. Hurricane Outage Prediction Model (HOPM)

This study uses the HOPM that was developed by Nateghi et al. (2014). It predicts the number of outages¹ in each grid cell. This prediction is the average of all the predictions from a large set of regression trees. The number of trees is determined by the *random Forest* function in R (Liaw and Weiner, 2002; R Core Team, 2013). The actual number of outages from each of the four hurricanes (Danny, Dennis, Ivan, and Katrina) are used as the response variable. The predictor variables that will be used in this study include hurricane characteristics (i.e., wind speed and duration of strongest winds), power system data (i.e., transformers, poles, overhead line, switches, customer density, and tree trimming), and environmental data. All of these datasets, with the exception of tree species, have been used in past versions of the HOPM (Han et al., 2009a,b; Quiring et al. 2011; Nateghi et al. 2014; Guikema et al., 2014).

Five different versions of the HOPM were generated for this study, each with different sets of predictor variables: All Variables, Public, Gust, Gust and Duration, and Reduced. The Reduced model uses the same variables as Nateghi et al. (2014). A full description of the predictor variables used in these models is provided in Table A1. All 5 versions of the HOPM were trained using the new tree species data. Training was done using a holdout analysis, where 30% of data were withheld, and the remainder of the data (70%) were used to train the model. The holdout analysis was repeated 30 times, and the 30% holdout was randomly generated for each

¹ Here an outage is defined as a “non-transitory activation of a protective device”. The protective devices include fuses, circuit breakers, or automatic circuit reclosers (Quiring et al. 2014). The nearest protective device upstream activates when an agent causes physical damage to the power system, such as when a tree falls on a line. This is done to isolate the damage, and all customers on that portion of the system temporarily lose power. This one incident is then considered to be a single outage and can result in varying numbers of customers losing power (Liu et al. 2007).

Table 2

Majority tree species in State A. The dominant (majority) tree species was determined for each grid cell. For example, Loblolly Pine is the dominant tree species in ~69% of the study region. The table only shows the tree species that were considered in this study.

Majority tree species	Percentage of study region
Loblolly Pine	69.20%
Sweetgum	11.17%
Other	4.74%
Empty	3.64%
Longleaf Pine	2.98%
Chestnut Oak	1.53%
Slash Pine	1.30%
Laurel Oak	1.02%
Water Oak	0.78%
White Oak	0.73%
Flowering Dogwood	0.58%
Virginia Pine	0.45%
TOTAL	98.11%

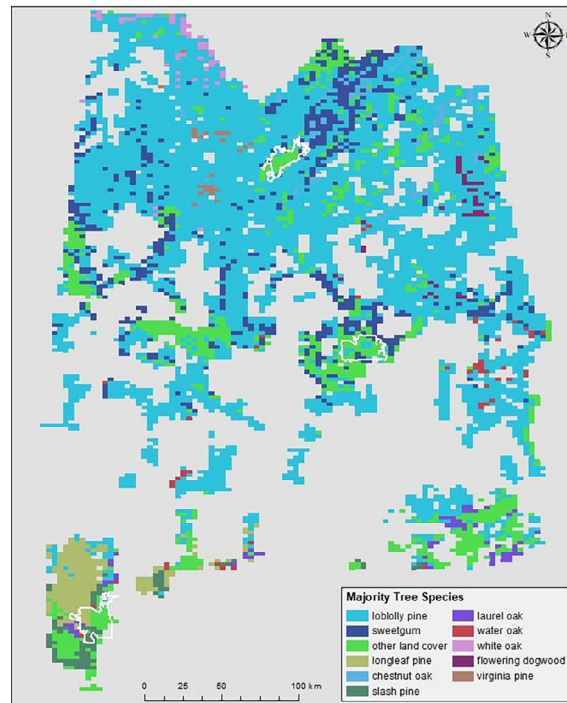


Fig. 2. Majority tree species in each grid cell. Urban areas are outlined in white and provided by the 2014 U.S. Census TIGER/Line Places shapefile, <https://www.census.gov/geo/maps-data/data/tiger-line.html>.

iteration. As in Nateghi et al. (2014), a mean-only, or “no model”, scenario is also to benchmark model performance. The mean-only prediction is generated by taking the mean of the response variable (actual outages) for the training data (70%) in each holdout. This scenario will serve as a baseline for each of the five models (i.e., it is the no-skill model).

After training is complete, the predictive accuracy of each model is assessed using the held-out data (30%). The *predict* function within the *randomForest* package in R is used to generate outage predictions for the withheld data. Both mean absolute error (MAE) and mean square error (MSE) were calculated for each of the thirty holdouts. The standard deviation from the mean (SD) is then computed for MAE and MSE using the *sd* function in R (R Core Team 2013). Because MSE gives more weight to the higher residuals (Guikema and Quiring, 2012), MSE is typically a better measure of error for normal distributions, while MAE is generally preferred for more uniform distributions (Chai and Draxler, 2014). However, Willmott and Matsuura (2005) argue that MAE should always be preferred over MSE in evaluating average model performance because the larger errors tend to have a greater influence on the total squared error than the smaller errors, resulting in increasingly larger values for MSE and root mean square error (RMSE, Willmott and Matsuura, 2005). In the evaluating these models, both MAE and RMSE are considered.

Finally, rather than describing the results from all five versions of the HOPM, for simplicity it was decided to first select the best model and then use that model to quantify the impact of tree species on model performance. The best model is selected using the accuracy assessment shown in Appendix A. The best model is that with the lowest MAE and MSE values. It also has the least spatial bias in absolute error (AE).

2.5. Local variable importance

Local variable importance (LVI) can describe the link between variables and samples (Touw et al., 2012). Each LVI score was calculated from the out-of-bag (OOB) data from all trees in the random forest. The OOB data comes from the validation data that is not used to train each tree. To calculate LVI, a portion of the OOB data is first permuted, or altered; then, the MSE for the permuted OOB data and the original OOB data are calculated. The permuted MSE is then subtracted from the original MSE to get a local importance score for each variable in each grid cell. The score is a measure of the predictive accuracy of the sample, and in this case, it is a measure of the impact on the correct prediction of outages. A negative score indicates a negative impact on the prediction of outages, or in other words, the variable did not improve the prediction. A 0 score is neutral, and a positive score indicates that the variable had a positive impact on, or improved, the prediction of outages. A higher score indicates a more positive impact on outage prediction (Grömping, 2009; Touw et al., 2012).

To calculate the LVI, scores were generated for each variable. As in Tonn et al. (2014), LVI scores were calculated for each grid cell by taking the mean importance of all four storms for the selected variable and then dividing that value by the sum of the mean importance scores for all of the variables. The result is a percentage importance value that indicates the importance, or contribution, of the variable in relation to the other variables in the model. LVI scores were only calculated for tree species having a significant relationship with outages, as indicated by the partial dependence plots and variable importance scores. An LVI map was generated to show the variable with the highest importance for each grid cell.

3. Results

3.1. Impact of tree species

The model selection process determined that the Reduced model (Nateghi et al. 2014) was the best model for this study. When the tree species predictor variables were added to the Reduced version of HOPM, error decreases slightly: MAE by 0.933% and MSE by only 6.854% (Table 3). This suggests that the addition of the tree species variables to HOPM increases model accuracy. Fig. 3a shows the polarized difference in AE by grid cell. Polarized differences were calculated by subtracting the AE of the original model without tree species from the AE of the model with the addition of tree species. A negative (–) value indicates an improvement in AE for that grid cell. Across the study area, there are some locations that show improvement in model performance and others that show a reduction in model performance (176 improved cells, 118 reduced), with the greatest differences occurring in and around urban areas (Fig. 3a), although the addition of tree species also has visible influences away from the most heavily populated regions as well. Table 4 shows the percentage differences between the improving and declining grid cells in Fig. 3a for each of the three standard deviation ranges. The differences are positive when more grid cells are improving with the addition of tree species, and negative when more grid cells see decreases in model performance when tree species are included. All ranges and the average are positive, so despite the small difference in improving and declining grid cells, more are improving, particularly in the higher ranges of standard deviation (Table 4).

Hot spot maps are also generated using the Getis-Ord G_i^* statistic (Getis and Ord, 1992; Ord and Getis, 1995), which uses spatial autocorrelation to identify statistically significant high and low clusters in the data (Fig. 3b). Regions in blue are where the model performs better (lower difference in AE), and those in red are where the model performs worse (higher difference in AE) with the addition of tree species. Multiple areas of model improvement are noted throughout the service territory, with the greatest improvement occurring in and around urban areas (Fig. 3b). There are also a few areas where model performance decreases when tree species are included, most notably in the north-central portion of the service area on the outskirts of a large city. These results could be related to the number of trees in urbanized areas, the presence of more overhead power lines in populated areas, and a greater

Table 3

Mean Absolute Error (MAE) and Mean Square Error (MSE) for the Reduced model with tree species (new) and without tree species (original). The third column features the percent change in MAE and MSE respectively from the original model to the new model with individual tree species. The MAE and MSE are also shown for the mean-only model. This is the null model that provides a baseline for comparison.

New MAE	0.747	– 0.933%
Original MAE	0.754	
New MSE	4.114	– 6.854%
Original MSE	4.406	
MAE (mean-only)	1.605	–
MSE (mean-only)	14.56	

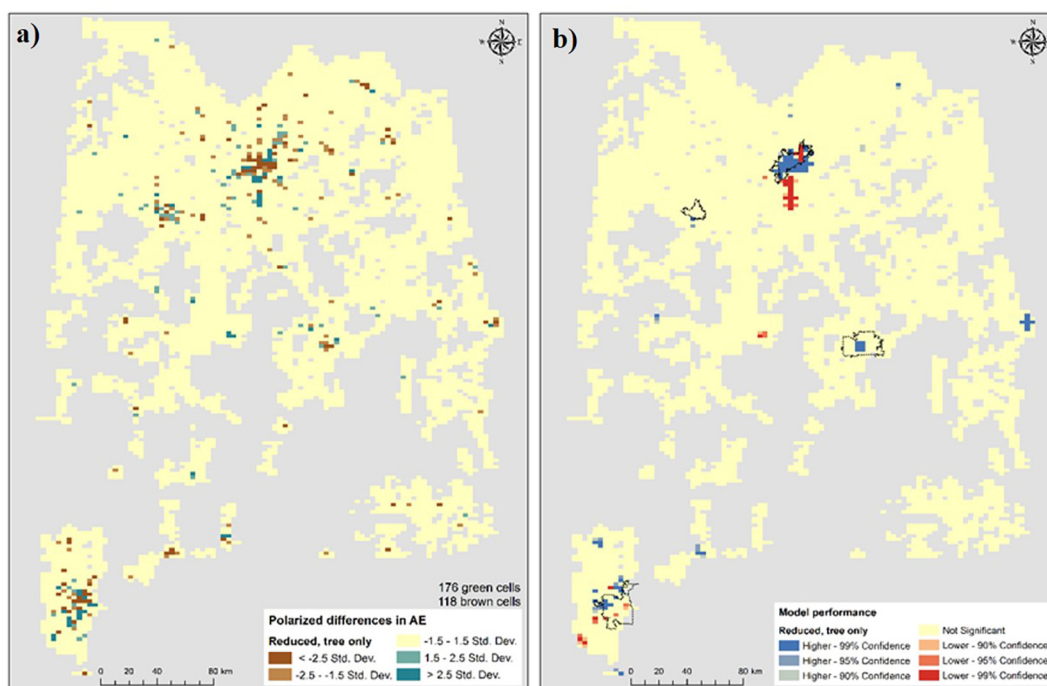


Fig. 3. a) Polarized differences in Absolute Error (AE) for Model 1 – Model 2. Model 1 is the Reduced model without tree species and Model 2 is the Reduced model with tree species. Differences > 1.5 Standard Deviation (SD) below the mean (green) indicate significant model improvement and differences less than -1.5 SD above the mean (brown) indicate significant model decline. b) Statistically significant clusters of positive (red) and negative (blue) polarized differences in AE for the Reduced model with the addition of tree species calculated using the Getis-Ord G_i^* statistic. More negative AE differences (blue) indicate higher model performance while more positive AE differences (red) indicate lower model performance. Urban areas are outlined in black. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4

Percent differences in model performance based on polarized differences in AE for each range of standard deviation (Fig. 3a).

± 0.5 to ± 1.5 SD	± 1.5 to ± 2.5 SD	$> \pm 2.5$	Average
7%	29%	37%	24%

proportion of trees that are susceptible to wind damage where there is a decline in model performance. However, the addition of tree species generally results in model improvement over more locations in the study region than those where model performance is reduced (Fig. 3b).

Table 5 shows the MAE for four outage groups as determined by the Natural Breaks (Jenks) classification scheme in ArcGIS (Jenks, 1967; Simpson and Human, 2008). MAE decreases for all outage groups, which indicates that the inclusion of tree variables increases the accuracy of the model. The decrease in model error ranges from $\sim 1\%$ to $\sim 5\%$ in these groups and the overall decrease in MAE is $\sim 3\%$. In order to test the statistical significance of these differences, a Wilcoxon signed rank test was performed. The Wilcoxon signed rank test is a non-parametric test procedure for the analysis of paired data (Woolson, 2008). The null hypothesis is that the differences have a distribution centered about zero. Based on the Wilcoxon test, the MAE values are significantly different at the 90% confidence interval (p -value = 0.0625).

Table 5

Change in Mean Absolute Error (MAE) and Standard Deviation (SD) by outage group. MAE (new) is for the Reduced model with the addition of tree species and MAE (original) is the Reduced model without tree species.

Actual outages	Number of cells	MAE (new)	MAE (old)	MAE (Δ)	SD
0–2	5805	0.264	0.274	–0.010	0.007
2–8	718	1.270	1.337	–0.067	0.047
8–21	134	3.245	3.333	–0.088	0.062
> 21	24	7.648	7.782	–0.134	0.095
Mean	–	3.107	3.182	–0.075	0.053

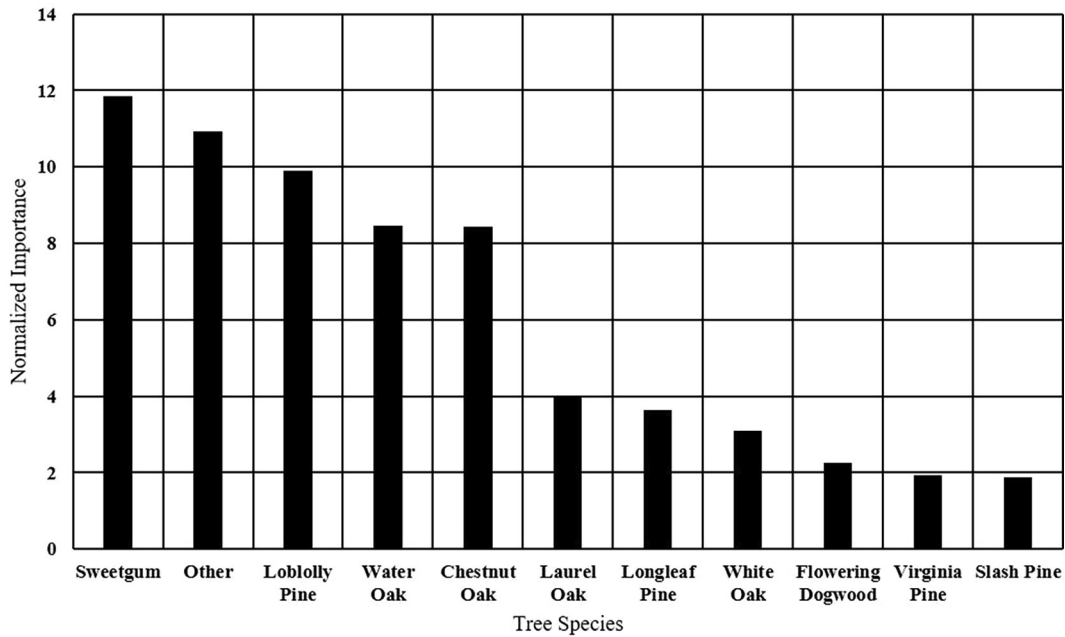


Fig. 4. Normalized variable importance for the each variable in the Reduced model. The maximum variable importance value is 100: Customer. Higher values indicate that a variable has greater importance in the Random Forest model.

Although the raw MAE numbers in Table 5 only show a modest increase in overall model accuracy, there are some much substantial improvements in performance at specific locations within the service territory (Fig. 3). Therefore, there are a number of potential advantages to including tree species data in HOPM. First, the inclusion of tree species variables can result in much larger local improvements in predictive accuracy. Second, identifying tree species that are associated with higher outage risk is important because utilities can use this information to target specific locations where these species are prevalent for more frequent tree trimming. Finally, trees are a major cause of power outages, therefore including more detailed information about trees makes physical sense and may allow for important effects to be captured by the models. For example, some utilities have reported increases in tree-related power outages due to the impacts of the Emerald Ash Borer and Southern Pine Beetle on tree health. The inclusion of tree species variables provides the potential to capture this non-meteorological risk factors that are associated with power outages.

3.2. Variable importance

Fig. 4 shows the normalized variable importance of all variables in the Reduced model. When normalized, the highest node purity is given a value of 100; as numbers decrease from 100, so does their relative importance to the model. The Customer and Trimming covariates had the highest impact on overall model performance, which generally agrees with Liu et al. (2008) and Nateghi et al. (2014). Sweetgum, Other, and Loblolly Pine each have normalized importance values greater than or around 10%. Water Oak and Chestnut Oak are also quite high, but less important than the first group of tree species (Fig. 4). Although the addition of tree species results in a statistically significant decrease in error, obviously these variables are less important than other factors such as the number of customers, frequency of tree trimming, windspeed and duration of strong winds. Therefore, tree species should be viewed as a secondary factor that can contribute to the accuracy of hurricane-related outage predictions. Our results suggest that the tree species variables can be locally important (i.e., specific grid cells), even though their overall contribution to model accuracy is modest.

Partial dependence plots were used to examine the nature of the relationship between each tree species and hurricane-related power outages. Partial dependence is a measure of the influence of a particular predictor variable on the response variable (in this case, outages), while the effects of other predictors are held constant (Nateghi et al., 2014). These partial dependence plots allow one to visualize the relationship between a particular tree species and outages. It can be useful for identifying whether the relationship is positive or negative and whether the relationship is non-linear or if thresholds exist.

The top 3 tree species in terms of variable importance are Sweetgum, Other and Loblolly Pine. Sweetgum has a direct linear relationship with outages (Fig. 5a). That is, as the relative abundance of Sweetgum increases, the number of outages also increase. This relationship holds between 0% and 70% relative abundance. Once the abundance is > 70%, the partial dependence line becomes horizontal which indicates that greater abundance of Sweetgum does not lead to more outages. Duryea et al. (2007) found that Sweetgum is one of the most wind-resistant species and that Sweetgum is less likely to break or be uprooted during hurricanes. Our findings seem to contradict those of Duryea et al. (2007). However, it is not possible for us to determine why higher relative abundance of Sweetgum trees are associated with more outages. It could be that Sweetgum trees may not be responsible for outages

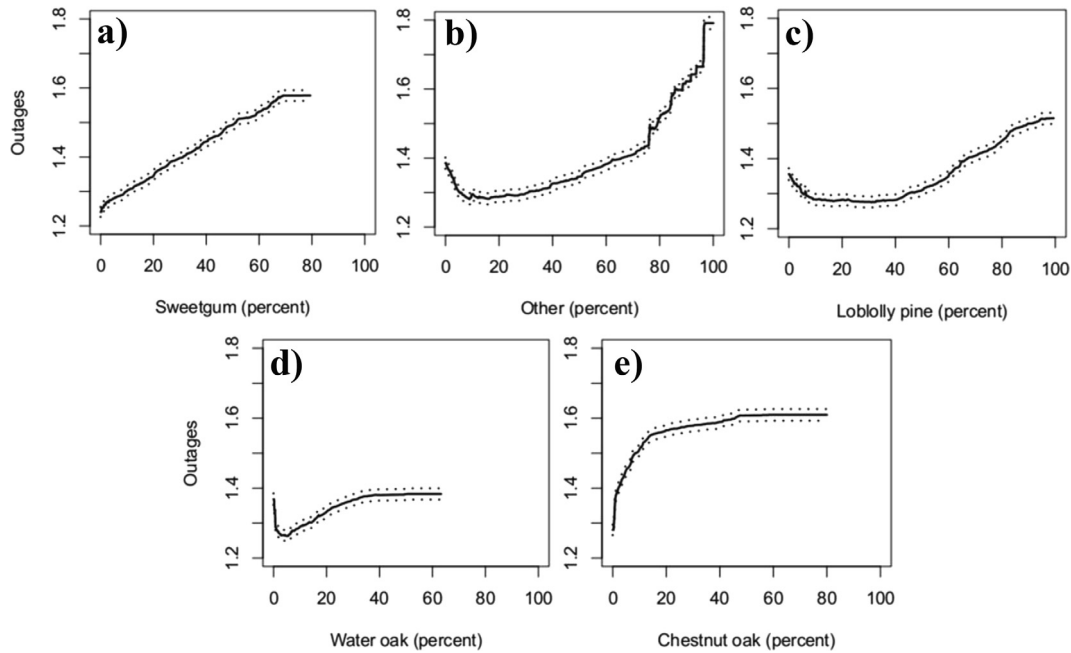


Fig. 5. Partial dependence plots showing the relative abundance of a given tree species within each grid cell versus the observed number of outages for: a) Sweetgum, b) Other, c) Loblolly Pine, d) Water Oak, and e) Chestnut Oak. For example, this shows that as the relative abundance of Sweetgum increases, the hurricane-related outages also increases.

in these areas and that they are serving as a proxy variable for another tree species (or another risk factor).

The second most important tree variable is the other category. It is associated with areas that have few trees such as farmland, urban regions, or water bodies. The partial dependence plot is substantially different from Sweetgum (Fig. 5b). It shows a non-linear relationship between the relative abundance of the other tree species and outages. The partial dependence is relatively flat below ~80%, while above ~80% prevalence, there is a strong positive association between outages and the other tree species. This category is most commonly associated with urban areas. These are the locations with the greatest number of customers and greatest concentration of electrical infrastructure. Quiring et al. (2011) previously found that including urban land cover variables in the HOPM lead to a statistically significant improvement in model performance. Therefore, it seems likely that the other tree species category is an important variable for the same reason. It indicates locations that are at greater risk for power outages, not necessarily because of trees breaking or being uprooted, but because they have a higher concentration of electrical infrastructure that is exposed to strong winds. However, a complete causal explanation for this pattern requires further investigation.

Loblolly Pine is the third most important tree species variable. It has a non-linear relationship with outages. At lower abundance (< 50%), the line is relatively flat which indicates that there is little relationship to outages. However, when the relative abundance of loblolly pine is > 60% there is a direct relationship with power outages (Fig. 5c). This finding agrees with Duryea et al. (2007) who found that Loblolly Pine may be a less wind-resistant species and that it only has about a 74% survival rate during hurricanes. In particular, loblolly pines are the species that are most likely to break during hurricanes and they are also relatively susceptible to uprooting.

Water Oak and Chestnut Oak are ranked fourth and fifth in terms of variables importance. Both of these tree species have similar partial dependence plots. As the relative abundance of Water Oak increases up to ~30%, the number of outages increases. Once the relative abundance of Water Oak is > 30% the partial dependence plateaus (Fig. 5d). This indicates that additional increases in the prevalence of Water Oak do not lead to an increased number of outages during hurricanes. Similarly for Chestnut Oak, there is a positive relationship between abundance and outages up to ~30% abundance and above that the line plateaus (Fig. 5e). This agrees with Duryea et al. (2007), which ranks Chestnut Oak, or swamp chestnut, as having a medium-high wind resistance, just below that of Sweetgum.

The remaining species in Table 1 are less important covariates and they have relatively little impact on the prediction of power outages. This includes Laurel Oak, Longleaf Pine, White Oak, Flowering Dogwood, Virginia Pine, and Slash Pine. In some cases, this may be because these tree species are not particularly abundant in the study region. All of these species are the dominant tree species in < 3% of the study region (Table 2). Longleaf Pine is the most prevalent and it is only the dominant tree species in 2.98% of the study region.

3.3. Local variable importance

The LVI was calculated for each tree species. This provides details about the spatial distribution of variable importance. The result

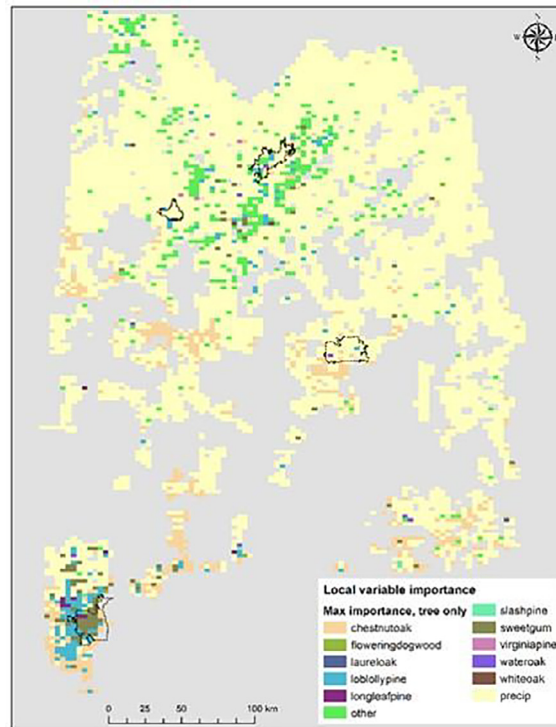


Fig. 6. Tree species with the highest Local Variable Importance (LVI, by %) in each grid cell.

is a percentage importance that indicates the contribution of that covariate with respect to the other covariates in the model. An LVI of zero indicates little or no importance, while positive LVI indicates that the covariate may help improve model performance in that location.

Fig. 6 shows the LVI for the tree species. The covariate with the highest LVI is shown for each grid cell. Chestnut Oak (beige), Loblolly Pine (sky blue), Sweetgum (hunter green), and the Other (spring green) covariates all seem to have the most widespread coverage (Fig. 6), and this pattern agrees with the partial dependence (Fig. 5) and LVI plots for each individual species (not shown).

Chestnut Oak (beige), Loblolly Pine (sky blue), and Sweetgum (hunter green) show more coherent spatial patterns (Fig. 6). This indicates that these species may have a significant effect on outages in these areas. For Sweetgum, the areas of highest importance tend to be urban regions, while Chestnut Oak are more widespread, affecting urban and rural regions almost equally. Chestnut Oak is important for explaining outages near the coast. The lack of spatial coherence for the other tree species suggests their influence may be more uniform across the service area. Tree species appear to have greater importance in the southwestern corner of the study region, where they are more important than almost all the other predictor variables.

4. Discussion and conclusions

This study has demonstrated that the inclusion of tree species data can result in a modest improvement in model accuracy. Although the addition of tree species results in a statistically significant decrease in error, these variables are much less important than other factors such as the number of customers, frequency of tree trimming, windspeed and duration of strong winds. Therefore, tree species should be viewed as a secondary factor that can contribute to the accuracy of hurricane-related outage predictions. Our results suggest that the tree species variables can be locally important (i.e., specific grid cells), even though their overall contribution to model accuracy is modest. Based on Fig. 6, it is evident that there is a relationship between tree species and outages in large portions of the study region. In the central part of the state, Chestnut Oak is important, while Sweetgum and Loblolly Pine are important in the southeastern part of the state and around urban areas. The accuracy of the outage predictions, especially near urban areas, where trees are likely to be present and maintained, improves when tree species are included in the model.

This study identified which tree species have the greatest impact on power outages using variable importance and partial dependence plots from State A. The most important tree species are Sweetgum, Loblolly Pine, Chestnut Oak, White Oak, and the Other category. This is likely partly related to the wind resistance of each species and partly related to other factors that are not accounted for in this study (i.e., the relative abundance of these species in the utility right of way, the age and health of the trees, vegetation management and trim cycles, etc.). For example, as the abundance of Sweetgum increased, this was found to cause a greater number of outages, even though, according to Duryea et al. (2007), it is a wind resistant species. On the other hand, Loblolly Pine has relatively low wind resistance and therefore, as expected, the number of outages increases as the abundance of Loblolly Pine

increases. Inclusion of tree species information can improve model performance and help to identify regions that are more susceptible to hurricane-related power outages in a given utility company's service territory. Information on tree species can also be used to optimize vegetation management and tree trimming cycles, rather than using the same trim cycle for the entire service territory. The identification of locations with susceptible tree species can be used to inform targeted preventative action, such as additional tree trimming, which is clearly a large contributor to outages in this HOPM study. Likewise, if a location contains tree species that are generally more wind resistant, they may require less frequent tree trimming.

Previous versions of HOPM have considered landcover data that were aggregated into eleven coherent land covers types: water, barren land, forest, scrub, grassland, hay/crops, wetlands, and four types of developed land based on the intensity of development. Most of the people and electrical infrastructure are located in developed areas, but the land cover does not provide any information about the number or type of trees in these developed areas. Therefore, this study allowed us to examine whether also including tree species data improved the model. As can be seen from our results, the inclusion of tree species results in a modest improvement in model performance, even when compared to the All Variables model which includes the land cover variables (Appendix A). This suggests that it may be worthwhile to include tree species data even in models that already include land cover.

There are a number of limitations to this study. First, the dominant tree species in each utility grid cell was used to represent all the trees. This does not account for the variety and heterogeneity of trees within each grid cell. Second, we were not able to identify the specific locations of overhead lines and poles. Therefore, we are assuming that the dominant tree species in each grid cell are also the tree species that are closest to the utility infrastructure. Third, this study is based on only one utility company and a small number of hurricanes. Fourth, the location and abundance of tree species is uneven in our study region. Loblolly Pine is the most abundant tree species in our study region and it was the dominant tree species in ~69% of the grid cells. Most of the other tree species are relatively rare (dominant in < 5% of grid cells) and this may mask their importance for predicting power outages. Therefore, future research should examine the relationships between tree species and power outages in other geographic regions (other utility service territories), using a larger number of storm events and considering only the trees that are located close enough to the electrical infrastructure to cause problems. This will help to better quantify the importance of including tree species information in power outage models and to identify the tree species that contribute to increased power outage risk. In addition, sampling strategies could be applied to deal with the imbalanced distribution of tree species. For example, Kabir et al. (in press) used resampling and cost-sensitive learning approaches to overcome the challenges associated with zero-inflated power outage data. Similarly, it would be possible to develop new datasets for retraining the HOPM using resampling methods to generate a more balanced sample of grid cells by over-sampling from the less common tree species and under-sampling from locations where Loblolly Pine is dominant. The advantage of this approach is that it would provide a better approach for isolating the influence and importance of each tree species.

Accurate and timely prediction of hurricane-related power outages will provide utility companies with an additional resource to compile either power restoration resources or tree trimming resources to prevent unnecessary expenditures or extended outage repair times. The prediction of power outages in advance of a land-falling hurricane has many benefits for utility companies, and potentially emergency managers and the general public as well. Utility companies generally lack sufficient personnel to efficiently restore power after a major event such as a hurricane, and they often must call on other neighboring utility providers for assistance. Prior to doing so, the utility company requesting assistance must first estimate the number of personnel and the amount of equipment that may be required to restore power quickly and effectively. A delicate balance must be maintained, as requesting more assistance than is needed could result in unnecessary costs, while requesting too little assistance may result in longer power restoration times. Thus, an accurate power outage prediction model, along with a knowledge of the key drivers of outages, will be beneficial for utility planning prior to such storm events.

Declaration of Competing Interest

None.

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Appendix A.: Model evaluation and selection

Six different versions of HOPM are tested with and without the addition of individual tree species as covariates. All of the variables tested in each version of HOPM are shown below in Table A1. Initially, Mean Absolute Error (MAE) and Mean Square Error (MSE) are calculated for each model with the addition of tree species. These values are shown in Table A.2. The two gust models (Gust and Gust + Duration) have the highest MAE and MSE, while the Public model (which only contains publically available data, i.e., nothing from the State A utility company) values are also somewhat lower (Table A.2).

Another method to visualize and compare error is through simple bar graphs. Fig. A.1 compares the MAE and MSE for all models before and after the inclusion of tree species and the Mean Only (No Model). The Public and All Variables models show little

Table A1

Predictor variables used in each of the five versions of the model and the Mean Only (No Model). New predictors not used in Nateghi et al. (2014) are bolded.

Version	Predictor Variables
All Variables	Elevation, Transformers, Poles, Overhead line, Switches, Time since last landfall, Wind speed (max), Duration of strongest winds, Slope, Aspect, Topography, Land cover type, Mean annual precipitation, Standardized Precipitation Index, Soil moisture (depth 1 ¹), Soil moisture (depth 3 ²), Soil type, Soil layers (number), Soil layer depth, Hydrologic group, Customer density, Tree trimming, Tree species
Public	Elevation, Time since last landfall, Wind speed (max), Duration of strongest winds, Slope, Aspect, Topography, Land cover type, Mean annual precipitation, Standardized Precipitation Index, Soil moisture (depth 1), Soil moisture (depth 3), Soil type, Soil layers (number), Soil layer depth, Hydrologic group, Customer density, Tree trimming, Tree species
Gust	Wind speed (max), Customer density, Tree species
Gust + Duration	Wind speed (max), Duration of strongest winds, Customer density, Tree species
Reduced	Wind speed (max), Duration of strongest winds, Soil moisture (depth 1), Soil moisture (depth 3), Customers (number), Tree trimming, Tree species
Mean Only (No Model)	Mean of the training (leftover) sample

¹ 0–10 cm soil depth.

² Soil depth below 40 m.

Table A2

Mean Absolute Error (MAE) and Mean Square Error (MSE) for each of the five models and the Mean Only (No Model). The Mean Only is only displayed for comparison and is not included in the calculation of Mean MAE/MSE or Mean SD.

	Reduced	All Variables	Public	Gust + Duration	Gust	Mean Only	Mean MAE/MSE	Mean SD
MAE	0.747	0.755	0.773	0.830	0.849	1.605	0.791	0.014
MSE	4.114	4.084	4.293	5.002	5.307	14.06	4.560	0.491

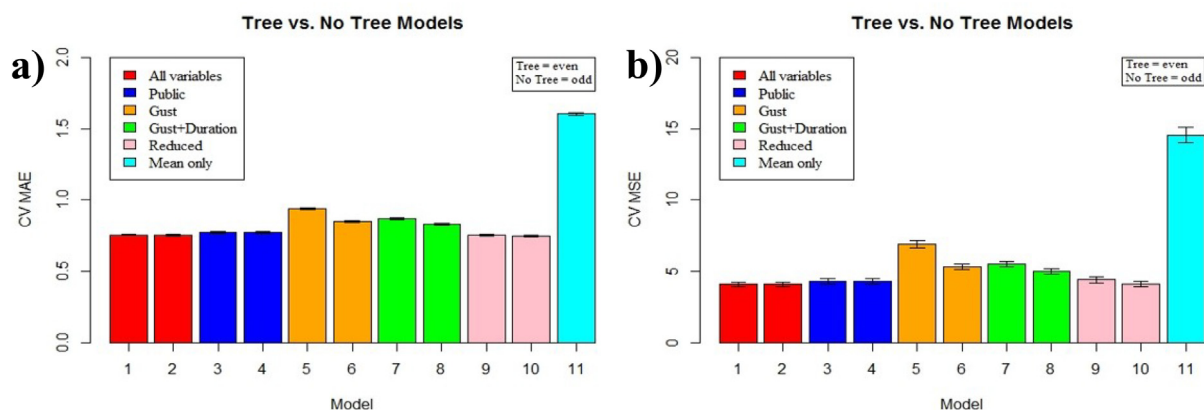


Fig. A1. a) Mean Absolute Error (MAE) and b) Mean Square Error (MSE) for each version of the HOPM, including the Mean Only (No Model). Odd numbers do not include tree species, while even numbers do. Lower values of MAE and MSE represent less error.

reduction in error after the inclusion of tree species due to the greater amount of variables in each. Error decreases in Gust, Gust + Duration, and the Reduced models upon the introduction of tree species. Despite the decreases, Gust and Gust + Duration still have very large errors relative to the other models (Fig. A.1), confirming that they will not be useful to the analysis of tree species. The Public model does not have a significant decrease in error and has higher error values than the All Variables and Reduced models (Table A.2, Fig. A.1). The two models with the lowest error values are the Reduced and All Variables models, broadly agreeing with those in Nateghi et al. (2014), so these two models will be used for the rest of the analysis in this Appendix.

Table A.3 shows the difference in errors between the Reduced and All Variables models with the tree species variables (new) and

Table A3

MAE and MSE of the models with tree species (New) and without tree species (Old) and the percent change (%C) between the New and Old models for the Reduced and All Variables models.

Model	MAE (New)	MAE (Old)	MAE (%C)	MSE (New)	MSE (Old)	MSE (%C)
Reduced	0.747	0.754	−0.928	4.114	4.406	−6.627
All Variables	0.755	0.757	−0.264	4.084	4.089	−0.122

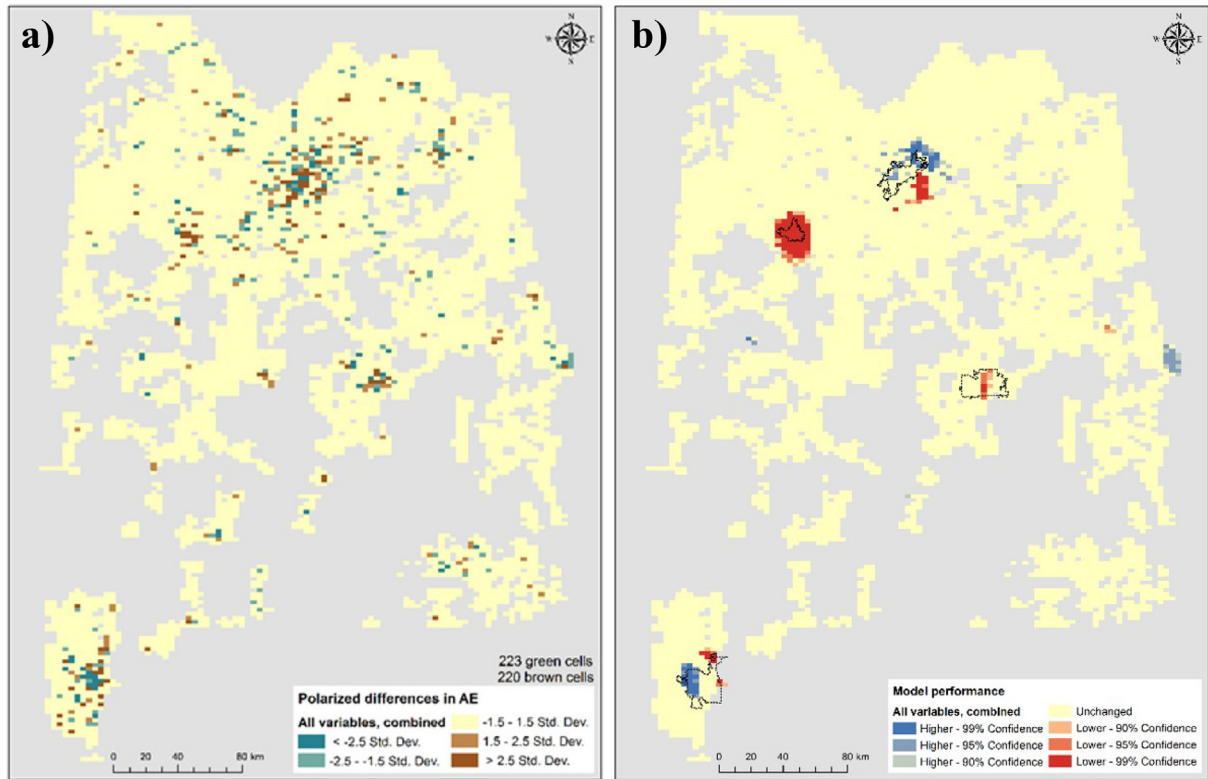


Fig. A2. a) Polarized differences in Absolute Error (AE) for Model 1 – Model 2. Model 1 is the All Variables model without tree species and Model 2 is the All Variables model with tree species. Differences > 1.5 Standard Deviation (SD) below the mean (green) indicate significant model improvement and differences less than -1.5 SD above the mean (brown) indicate significant model decline, b) Statistically significant clusters of positive (red) and negative (blue) polarized differences in AE for the All Variables model with the addition of tree species calculated using the Getis-Ord G_i^* statistic. More negative AE differences (blue) indicate higher model performance while more positive AE differences (red) indicate lower model performance. Urban areas are outlined in black.

without the tree species variable (old) and the percent change (%C) as calculated in Equation (A.1):

$$\%C = \frac{\bar{error}_b - \bar{error}_a}{\bar{error}_a} \times 100\% \quad (A.1)$$

where b is the error in the models that include tree species variables (new) and a is the error in the models without tree species (old). %C will be calculated the Reduced and All Variables models. A negative value for %C indicates a decrease in error when tree species are included, while a positive value indicates an increase in error.

The Reduced model has much larger reductions in error when adding the tree species variables (Fig. A.1, Table A.3), particularly MSE, while adding tree species to the All Variables model has very little influence on the model error (although %C is still negative, meaning the addition of tree species still improves the All Variables model). With the much larger number of covariates in the All Variables model (Table A.1), it makes sense that the addition of one addition covariate will have little impact on the reduction or increase of error. Additionally, the All Variables model already contains a Land cover variable, which may ultimately serve as a proxy for individual tree species. This would require further analysis and testing beyond the original intentions and scope of this current project.

It is also beneficial to view error spatially. Fig. A.2a shows the polarized differences in Absolute Error (AE) between Model 1, which does not include tree species (Old), and Model 2, which does include individual tree species (New) for the All Variables models. This figure can be compared to Fig. 3a. The All Variables model with tree species features 223 cells with improvement, greater than the Reduced model, but also features 220 cells with a decline in model performance, much higher than the 118 cells of model decline in the Reduced model (compare Figs. A.2a and 3a). Fig. A.2b shows the model performance with statistically significant clusters of poorer or greater model performance in polarized differences using the Getis-Ord G_i^* statistic in ArcGIS for the All Variables model. When comparing Figs. A.2b to 3b, it is clear that the All Variables model features a much larger portion of the study area with declining model performance including the entirety of a northwestern city.

Of the 5 models and the Mean Only (No Model), the Reduced model shows the most improvement when tree species are added. The All Variables and Public models perform well, but might be masking the actual impact of tree species due to their large amount of variables. Additionally, Nateghi et al. (2014) used the Reduced model for their study, so this will make an easier basis for comparison.

For these reasons, and the reduced amount of time necessary to run the Reduced model, the Reduced version of HOPM is used for analysis of added tree species on model performance.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.crm.2019.100193>.

References

- Barry, P.J.C. Doggett, R.L. Anderson, K.M.S. Swain, 1993. How to Evaluate and Manage Storm-Damaged Forest Areas. USDA Forest Service Southern Region Management Bulletin R8-MB 63, pp. 18.
- Beven, J., 2005. Tropical Cyclone Report: Hurricane Dennis, 4-13 July 2005. Available online at https://www.nhc.noaa.gov/data/tcr/AL042005_Dennis.pdf.
- Chai, T., Draxler, R.R., 2014. Root mean square error (RMSE) or mean absolute error (MAE)?—Arguments against avoiding RMSE in the literature. *Geosci. Model Dev.* 7, 1247–1250. <https://doi.org/10.5194/gmd-7-1247-2014>.
- Chang, S.E.T. McDaniels, D.A. Reed, 2004. Mitigation of extreme events: Electric power outages and cascading effects. Proceedings of USC Center for Risk and Economic Analysis of Terrorism Events (CREATE) Symp. on the Economic Cost and Consequences of Terrorism, CREATE, Los Angeles, CA.
- Davidson, R.A., Liu, H., Sarpong, I.K., Sparks, P., Rosowsky, D.V., 2003. Electric power distribution system performance in Carolina hurricanes. *Nat. Hazards Rev.* 4, 36–45.
- Duryea, M.L., Kampf, E., Littell, R.C., 2007. Hurricanes and the urban forest: I. Effects on southeastern United States coastal plain tree species. *Arboric. Urban For.* 33 (2), 83–97.
- Emanuel, K.A., 2005. *Divine Wind: The History and Science of Hurricanes*. Oxford University Press, New York, pp. 285.
- Getis, A., Ord, J.K., 1992. The analysis of spatial association by distance statistics. *Geogr. Anal.* 24 (3), 189–206.
- Gresham, C.A., Williams, T.M., Lipscomb, D.J., 1991. Hurricane Hugo wind damage to southeastern US coastal forest tree species. *Biotropica* 23 (4), 420–426. <https://doi.org/10.2307/2388261>.
- Grömping, U., 2009. Variable importance assessment in regression: Linear regression versus random forest. *Am. Stat.* 63 (4), 308–319. <https://doi.org/10.1198/tast.2009.08199>.
- Guikema, S.D., Davidson, R.A., Liu, H., 2006. Statistical models of the effects of tree trimming on power system outages. *IEEE Trans. Power Activity* 21, 1549–1557.
- Guikema, S.D., Quiring, S.M., Han, S.R., 2010. Prestorm estimation of hurricane damage to electric power distribution systems. *Risk Anal.* 30, 1744–1752.
- Guikema, S.D., Quiring, S.M., 2012. Hybrid data mining-regression for infrastructure risk assessment based on zero-inflated data. *Reliab. Eng. Sys. Saf.* 99, 178–182. <https://doi.org/10.1016/j.res.2011.10.012>.
- Guikema, S.D., Nateghi, R., Quiring, S.M., Staid, A., Reilly, A.C., Gao, M., 2014. Predicting hurricane power outages to support storm response planning. *IEEE Access* 2, 1364–1373. <https://doi.org/10.1109/ACCESS.2014.2365716>.
- Han, S.R., Guikema, S.D., Quiring, S.M., 2009a. Improving the predictive accuracy of hurricane power outage forecasts using generalized additive models. *Risk Anal.* 29, 1443–1453.
- Han, S.R., Guikema, S.D., Quiring, S.M., Lee, K.H., Rosowsky, D., Davidson, R.A., 2009b. Estimating the spatial distribution of power outages during hurricanes in the Gulf coast region. *Reliab. Eng. Syst. Saf.* 94, 199–210. <https://doi.org/10.1016/j.res.2008.02.018>.
- Jenks, G.F., 1967. The data model concept in statistical mapping. *Int. Yearbook Cartogr.* 7, 186–190.
- Kabir, E., Guikema, S. D., and S. M. Quiring, (in press): Probabilistic mixture models for predicting thunderstorm-induced power outages. *IEEE Trans. Power Syst.* DOI: 10.1109/TPWRS.2019.2914214.
- Knabb, R. D., J. R. Rhone, and D. P. Brown, 2005: Tropical Cyclone Report: Hurricane Katrina, 23-30 August 2005. [Available online at https://www.nhc.noaa.gov/data/tcr/AL122005_Katrina.pdf].
- Krishnamurthy, V., Kwasinski, A., 2013. Characterization of power system outages caused by hurricanes through localized intensity indices. *Power and Energy Society General Meeting (PES), 2013 IEEE*. IEEE 1–5.
- Krist Jr., F. J. et al., 2014: 2013-2027 National Insect and Disease Forest Risk Assessment. USDA Forest Service, Forest Health Technology Enterprise Team, pp. 209.
- Liaw, A., Weiner, M., 2002. Classification and regression by randomForest. *R News* 2 (3), 18–22.
- Liu, H.B., Davidson, R.A., Rosowsky, D.V., Stedinger, J.R., 2005. Negative binomial regression of electric power outages in hurricanes. *J. Infrastruct. Syst.* 11, 258–267.
- Liu, H., Davidson, R.A., Apanasovich, T., 2007. Statistical forecasting of electric power restoration times in hurricanes and ice storms. *IEEE Trans. Power Syst.* 22, 2270–2279. <https://doi.org/10.1109/TPWRS.2007.907587>.
- Liu, H., Davidson, R.A., Apanasovich, T.V., 2008. Spatial generalized linear mixed models of electric power outages due to hurricanes and ice storms. *Reliab. Eng. Syst. Saf.* 93, 897–912. <https://doi.org/10.1016/j.res.2007.03.038>.
- Maliszewski, P.J., Larson, E.K., Perrings, C., 2012. Environmental determinants of unscheduled residential outages in the electrical power distribution of Phoenix, Arizona. *Reliab. Eng. Sys. Safety* 99, 161–171.
- Nateghi, R., Guikema, S., Quiring, S.M., 2014. Power outage estimation for tropical cyclones: improved accuracy with simpler models. *Risk Anal.* 34, 1069–1078. <https://doi.org/10.1111/risa.12131>.
- Nojima, N. Sugito, M., 2002. Empirical estimation of outage and duration of lifeline disruption due to earthquake disaster. Proceedings of US-China-Japan Workshop on Lifeline Systems, Qindao, China.
- Ord, J.K., Getis, A., 1995. Local spatial autocorrelation statistics: distributional issues and an application. *Geogr. Anal.* 27 (4), 286–306.
- Ouyang, M., Dueñas-Osorio, L., 2014. Multi-dimensional hurricane resilience assessment of electric power systems. *Struct. Saf.* 48, 15–24. <https://doi.org/10.1016/j.strusafe.2014.01.001>.
- Pasch, R.J., 1997: Preliminary Report: Hurricane Danny, 16-26 July 1997. [Available online at http://origin-www.nhc.noaa.gov/data/tcr/AL051997_Danny.pdf].
- Quiring, S.M., Zhu, L.Y., Guikema, S.D., 2011. Importance of soil and elevation characteristics for modeling hurricane-induced power outages. *Nat. Hazards* 58 (1), 365–390. <https://doi.org/10.1007/s11069-010-9672-9>.
- Quiring, S.M., Schumacher, A.B., Guikema, S.D., 2014. Incorporating hurricane forecast uncertainty into a decision-support application for power outage modeling. *Bull. Amer. Meteor. Soc.* 95, 47–58. <https://doi.org/10.1175/BAMS-D-12-00012.1>.
- R Core Team, 2013: R: A language and environment for statistical computing. (Available online at <https://www.r-project.org/>).
- Reed, D.A., 2008. Electric utility distribution analysis for extreme winds. *J. Wind Eng. Ind. Aerodyn.* 96, 123–140.
- Reed, D.A., Powell, M.D., Westerman, J.M., 2010a. Energy supply system performance for Hurricane Katrina. *J. Ener. Eng.* 136 (4), 95–102. [https://doi.org/10.1061/\(ASCE\)EY.1943-7897.0000028](https://doi.org/10.1061/(ASCE)EY.1943-7897.0000028).
- Reed, D.A., Powell, M.D., Westerman, J.M., 2010b. Energy infrastructure damage analysis for Hurricane Rita. *Nat. Hazards Rev.* 11, 102–109.
- Ruefenacht, B. et al., 2008. Conterminous U.S. and Alaska Forest Type Mapping Using Forest Inventory and Analysis Data. (Available online at http://data.fs.usda.gov/geodata/rastergateway/forest_type/).
- Schott, T., Landsea, C. Hafele, G. Lorens, J. Taylor, A. Thurm, H. Ward, B. Willis, M. Zaleski, W. 2012. The Saffir-Simpson hurricane wind scale. (Available online at <https://www.nhc.noaa.gov/pdf/sshws.pdf>).
- Simpson, P., Van Bossuyt, R., 1996. Tree-caused electric outages. *J. Arboric.* 22, 117–121.
- Simpson, D.M., Human, R.J., 2008. Large-scale vulnerability assessments for natural hazards. *Nat. Hazards* 47 (2), 143–155.

- Stewart, S. R., 2004: Tropical Cyclone Report: Hurricane Ivan, 2-24 September 2004. (Available online at https://www.nhc.noaa.gov/data/tcr/AL092004_Ivan.pdf).
- Tonn, G., Guikema, S., Ferreira, C., Quiring, S., 2014. A longitudinal analysis of the drivers of power outages during hurricanes: A case study with Hurricane Isaac. Working Paper, Probabilistic Safety Assessment and Management (PSAM), Johns Hopkins University, Department of Geography & Environmental Engineering, Baltimore, MD.
- Touw, W.G., Bayjanov, J.R., Overmars, L., Backus, L., Boekhorst, J., Wels, M., van Hijum, S.A.F.T., 2012. Data mining in the Life Sciences with Random Forest: a walk in the park or lost in the jungle? *Briefings Bioinf.* 14 (3), 315–326. <https://doi.org/10.1093/bib/bbs034>.
- Vanderford, M.L., Nastoff, T., Telfer, J.L., Bonzo, S.E., 2007. Emergency communication challenges in response to Hurricane Katrina: lessons from the centers for disease control and prevention. *J. Appl. Comm. Res.* 35 (1), 9–25. <https://doi.org/10.1080/00909880601065649>.
- Willmott, C.J., Matsuura, K., 2005. Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance. *Clim. Res.* 30, 79–82.
- Winkler, J., Dueñas-Osorio, L., Stein, R., Subramanian, D., 2010. Performance assessment of topologically diverse power systems subjected to hurricane events. *Reliab. Eng. Syst. Saf.* 95, 323–336. <https://doi.org/10.1016/j.ress.2009.11.002>.
- Woolson, R.F., 2008. Wilcoxon signed-rank test *Wiley Encyclopedia of Clinical Trials*. John Wiley & Sons Inc doi: 10.1002/9780471462422.eoct979.
- Xi, W., Peet, R.K., 2008. Hurricane effects on the Piedmont forests: patterns and implications. *Ecol. Restor.* 26 (4), 295–298.